A Multi-state Modeling Method to predict 10-year Disease trajectories between Coronary Heart Disease and Heart Failure

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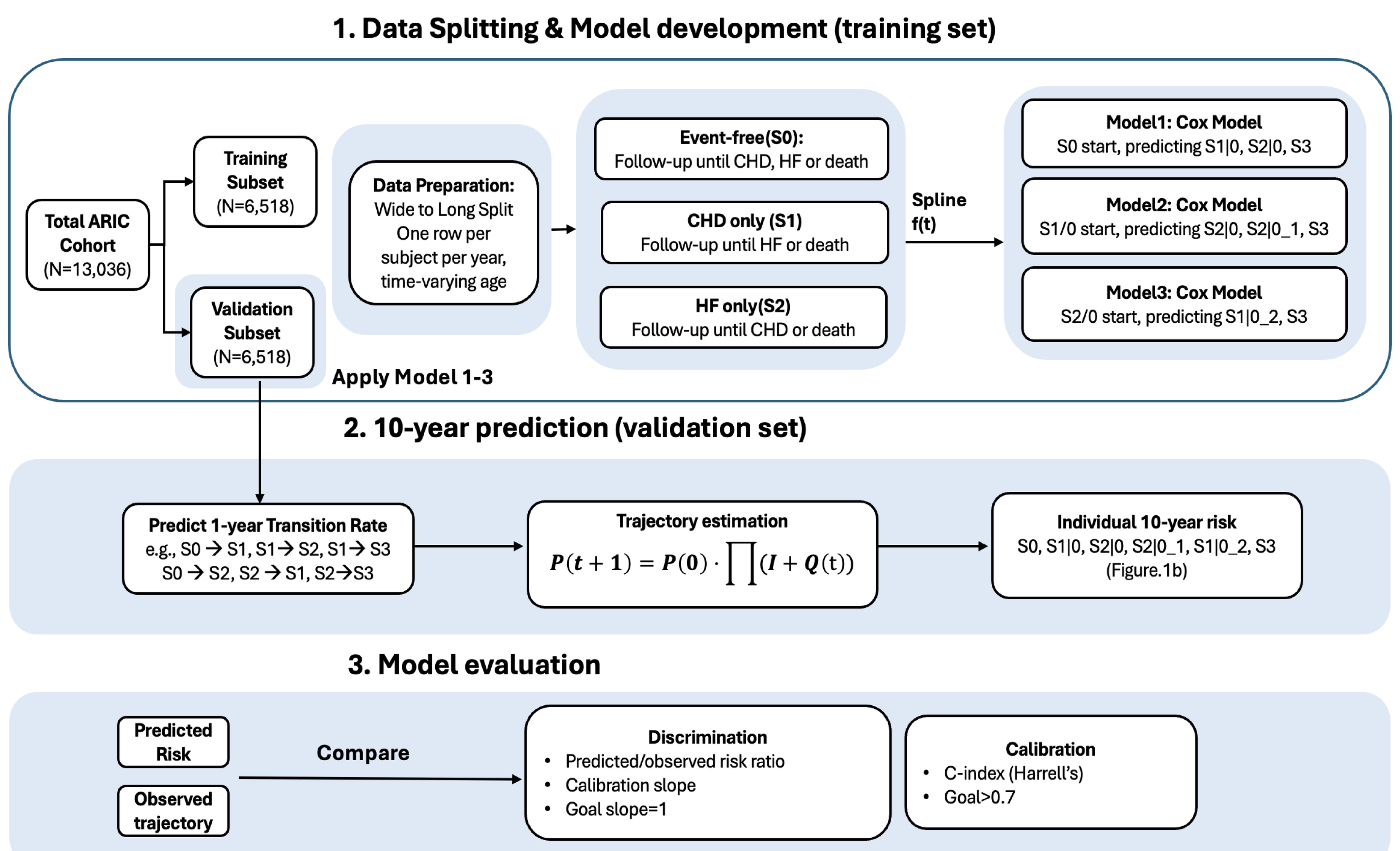
Background

Traditional cardiovascular risk models (e.g., PREVENT) only predict the **first event**, such as coronary heart disease (CHD) or heart failure (HF). However, patients often experience **multiple sequential cardiovascular events**, and these disease pathways have different prognoses and treatment strategies. Ignoring disease trajectories may underestimate disease complexity and limit personalized risk prediction.

Objective

To develop a multi-state model predicting an individual's 10-year risk of following each of six possible health trajectories after coronary heart disease (CHD) and heart failure (HF).

Methodology



Formula for step 1 models:

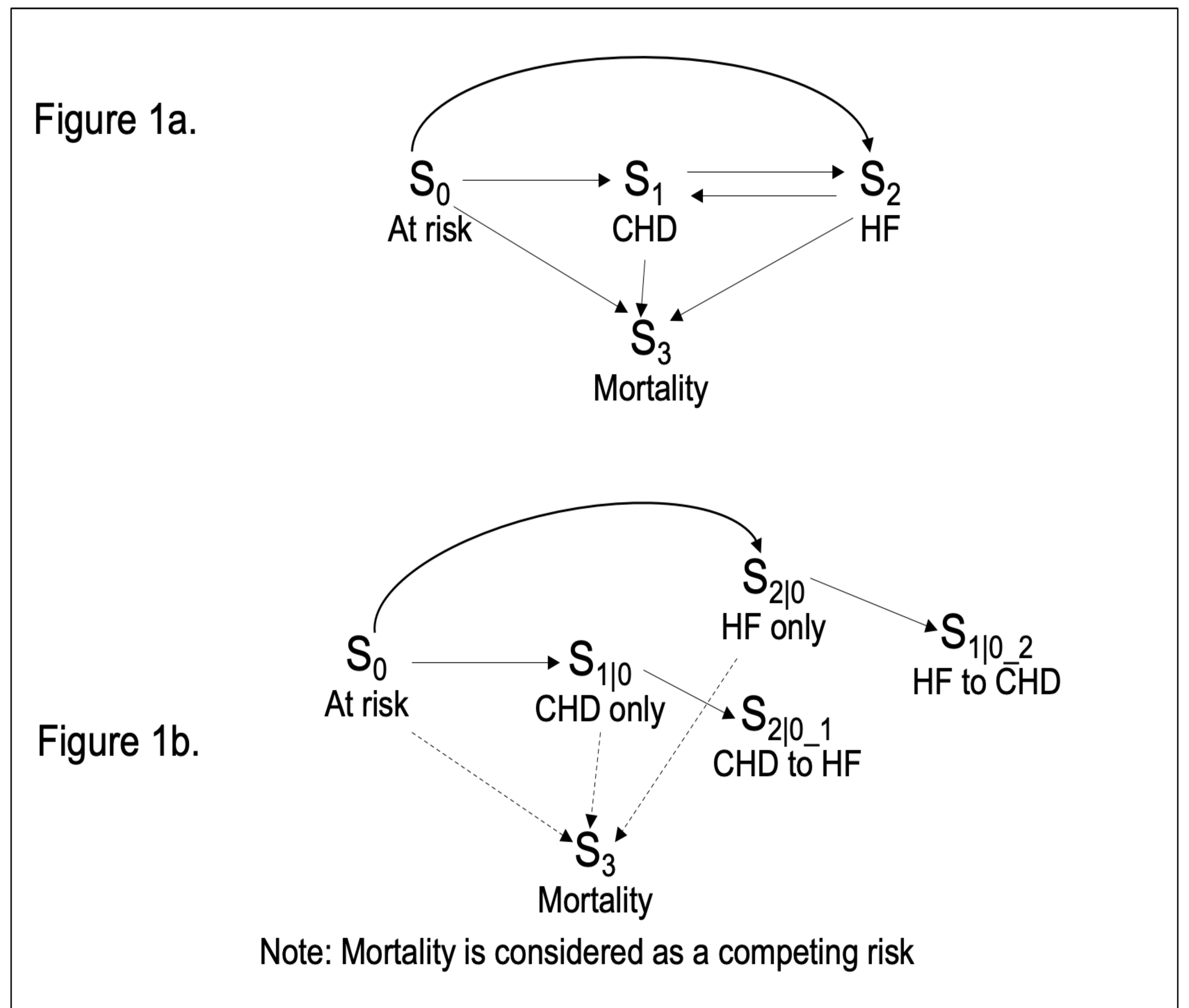
Model 1 starts from S_0: $h_{(S_m=1|S_0=1)}(\tau, t, \boldsymbol{V}) = h_{0m}(\tau)\exp(f(t) + \boldsymbol{V})$, m = 1, 2, 3.

Model 2 starts from S_1: $h_{(S_m=1|S_1=1,S_0=1)}(\tau, t, \boldsymbol{V}) = h_{1m}(\tau)\exp(f(t) + \boldsymbol{V})$, m = 2, 3.

Model 3 starts from S_2: $h_{(S_m=1|S_2=1,S_0=1)}(\tau, t, J_{0\_1}, \boldsymbol{V}) = h_{2m}(\tau)\exp(f(t) + \boldsymbol{V})$, m = 1, 3.

where $f(t)$ is a function of age, τ is time to event, $h_{0m}(\tau)$ is the baseline hazard of developing state m. V denotes the baseline predictors: sex, race, diabetes, smoking, blood pressure, lipids, eGFR, medication use.

Results



Characteristics	ARIC
Number of participants (N)	13,036
Age, year	54 (49, 59)
Female, %	7,192 (55.2%)
Blacks, %	3,218 (24.7%)
Total cholesterol, mmol/L	212 (186, 239)
HDL cholesterol, mmol/L	49.1 (39.5, 61.6)
SBP, mmHg	130 (112, 152)
BMI, kg/m^2	26.7 (23.9, 30.2)
eGFR, mL/min per 1.73 m^2	103.2 (95, 111.4)
Diabetes, %	1,134 (8.7%)
Current Smokers, %	3,393 (26.0%)
BP medication use, %	2,837 (21.8%)
Statin use, %	60 (0.5%)
Follow-up time, year	28 (19.8, 29.2)

Table 1. Baseline Characteristics of Participants in the Ath aerosclerosis Risk in Communities (ARIC) Study. The ARICa data were obtained from the NHLBI BioLINCC under a Data Use Agreement (DUA).

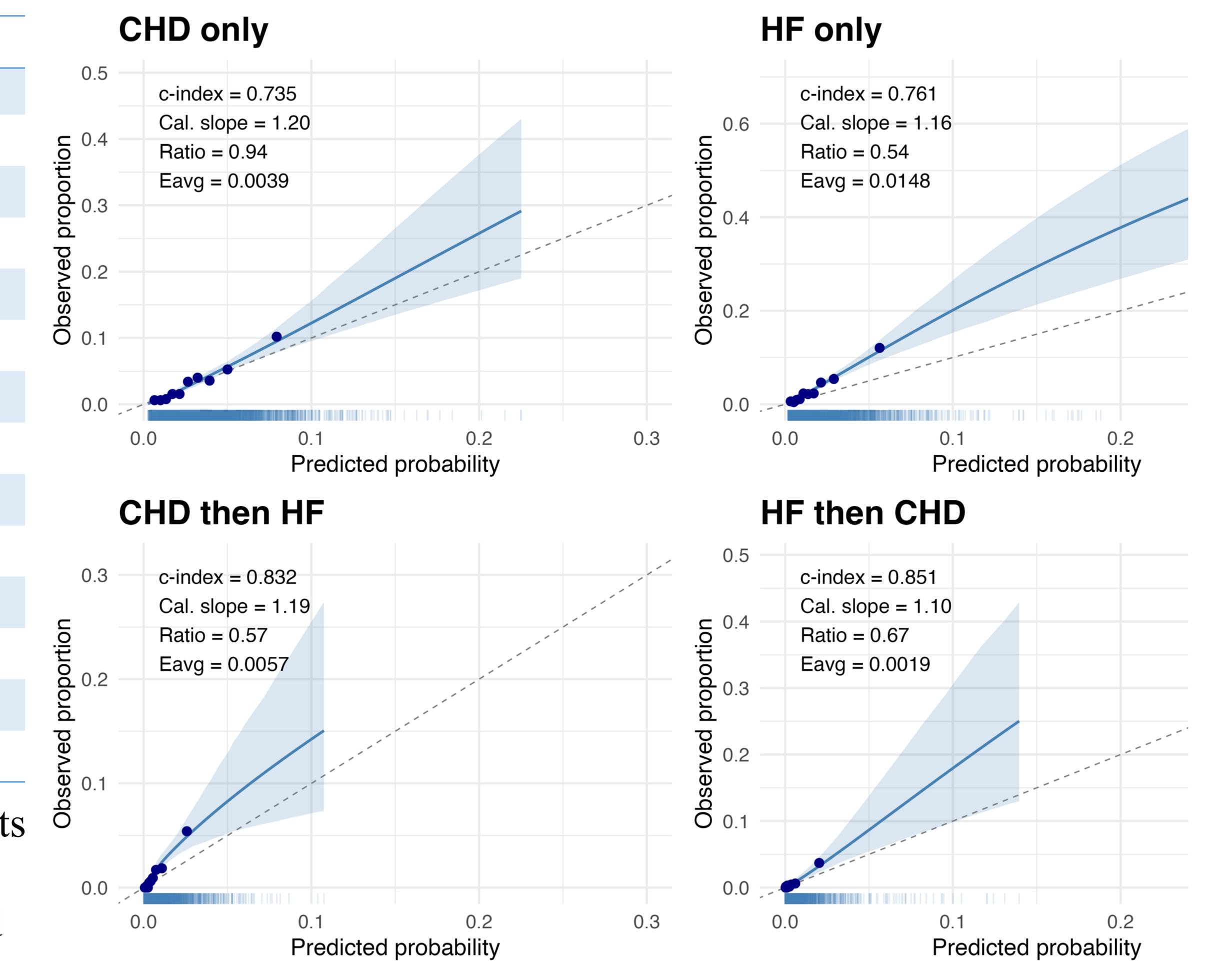


Figure 3. Calibration of 10-year predicted disease path

	S0→S1 n=1.5k/13k	S0→S2 n=2.2k/13k	S1→S2 n=654/1.5k	S2→S1 n=410/2.2k
Age (per year)	**1.05 (1.04, 1.06)**	**1.10 (1.09, 1.11)**	**1.05 (1.03, 1.06)**	**1.03 (1.01, 1.05)**
Sex (male vs. female)	**1.82 (1.62, 2.04)**	**1.21 (1.10, 1.33)**	**0.76 (0.64, 0.91)**	1.12 (0.91, 1.39)
Race (black vs. non-black)	1.11 (0.98, 1.27)	**1.41 (1.27, 1.56)**	**1.29 (1.06, 1.56)**	**1.46 (1.17, 1.81)**
Total cholesterol (per SD)	**1.29 (1.23, 1.35)**	1.01 (0.97, 1.06)	0.96 (0.89, 1.03)	**1.21 (1.11, 1.32)**
HDL cholesterol (per SD)	**0.75 (0.68, 0.82)**	0.97 (0.90, 1.04)	0.95 (0.81, 1.10)	0.89 (0.75, 1.07)
SBP (per SD)	0.96 (0.89, 1.05)	1.01 (0.95, 1.09)	1.07 (0.93, 1.22)	0.98 (0.83, 1.15)
BMI (per SD)	**1.07 (1.01, 1.14)**	**1.37 (1.31, 1.43)**	**1.22 (1.13, 1.33)**	0.91 (0.82, 1.00)
eGFR (per SD)	**0.91 (0.86, 0.97)**	**0.91 (0.87, 0.95)**	1.00 (0.92, 1.09)	**0.89 (0.83, 0.96)**
Diabetes (yes vs. no)	**2.23 (1.93, 2.59)**	**2.35 (2.09, 2.64)**	**1.60 (1.32, 1.95)**	**2.05 (1.63, 2.57)**
Current smoking (yes vs. no)	**1.91 (1.71, 2.14)**	**2.32 (2.11, 2.55)**	**1.25 (1.05, 1.49)**	**1.43 (1.16, 1.78)**
BP med. use (yes vs. no)	**1.42 (1.26, 1.60)**	**1.45 (1.32, 1.59)**	1.13 (0.95, 1.35)	1.19 (0.96, 1.46)
Statin use (yes vs. no)	0.73 (0.32, 1.62)	1.01 (0.57, 1.78)	0.88 (0.22, 3.55)	1.52 (0.56, 4.11)

2.35 (higher risk) – 1.00 – 0.73 (lower risk)

Figure 2. Multivariable-adjusted associations of baseline predictors with hazards of transition between states

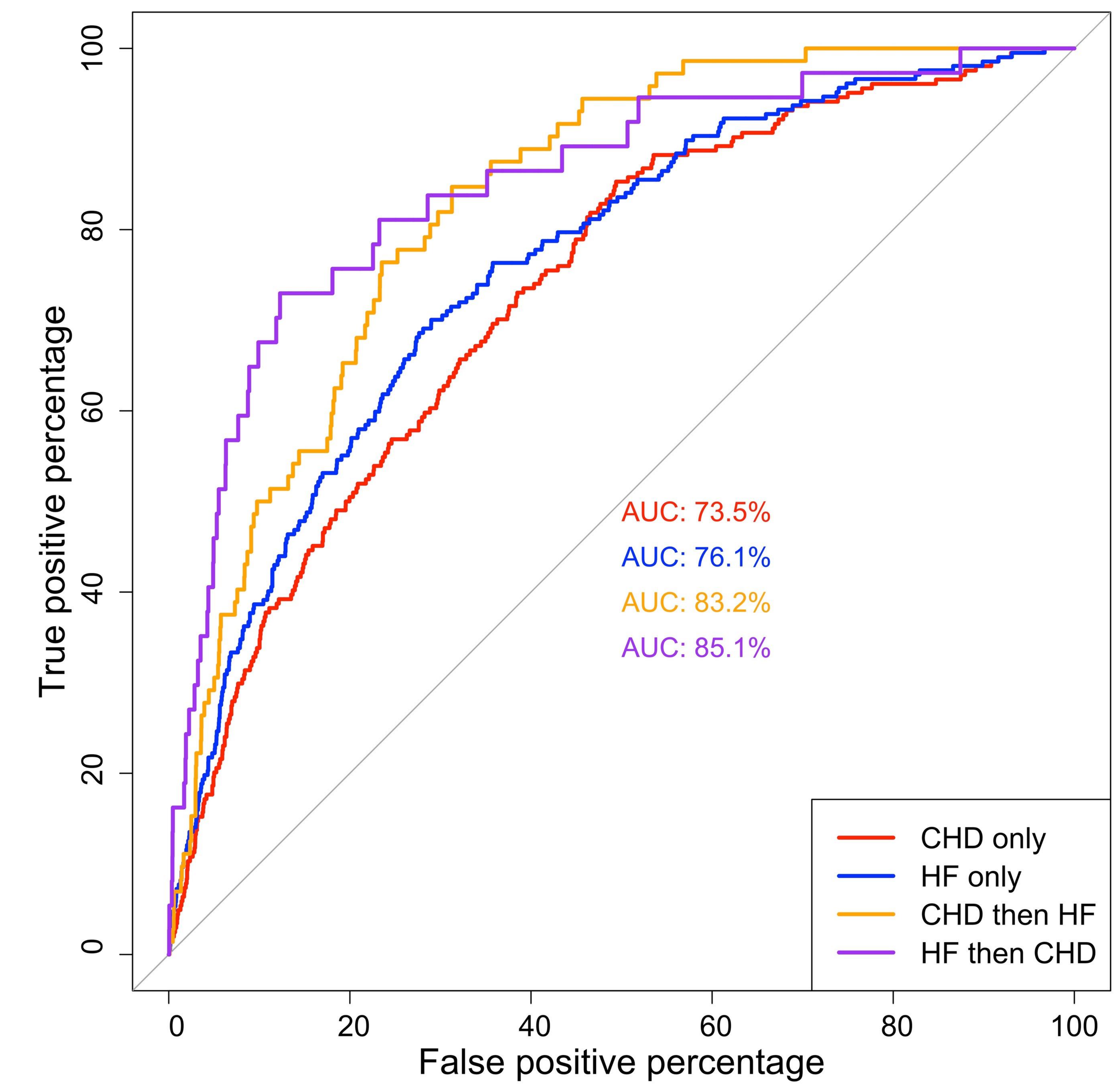


Figure 4. ROC Curves for 10-year predicted disease path

	CHD only	HF only	CHD—HF path	HF—CHD path
Women				
Observed cases	69	105	32	16
c-index	0.73 (0.73, 0.79)	0.77 (0.76, 0.79)	0.86 (0.84, 0.88)	0.93 (0.91, 0.95)
Predicted risk	1.91%	1.70%	0.49%	0.37%
Observed risk	1.92%	2.92%	0.89%	0.45%
Calibration slope	1.21	1.10	1.18	1.34
Men				
Observed cases	135	102	40	21
c-index	0.68(0.67, 0.70)	0.75(0.73, 0.77)	0.80(0.78, 0.82)	0.78(0.74, 0.82)
Predicted risk	4.26%	1.76%	0.81%	0.40%
Observed risk	4.69%	3.55%	1.39%	0.73%
Calibration slope	1.22	1.29	1.21	0.90

Table 2. 10-year ROC and calibration for women and men based on the CSC model including traditional risk factors.

Conclusion

Unlike single-endpoint tools such as PREVENT, this framework distinguishes *which* trajectory a person is likely to follow, not just *whether* an event occurs, with good discrimination and calibration.

Future Development

- External validation in independent cohorts to test generalizability across populations.
- Incorporation of time-varying predictors, updated during follow-up rather than fixed at baseline.

References

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